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Link Layer and LAN

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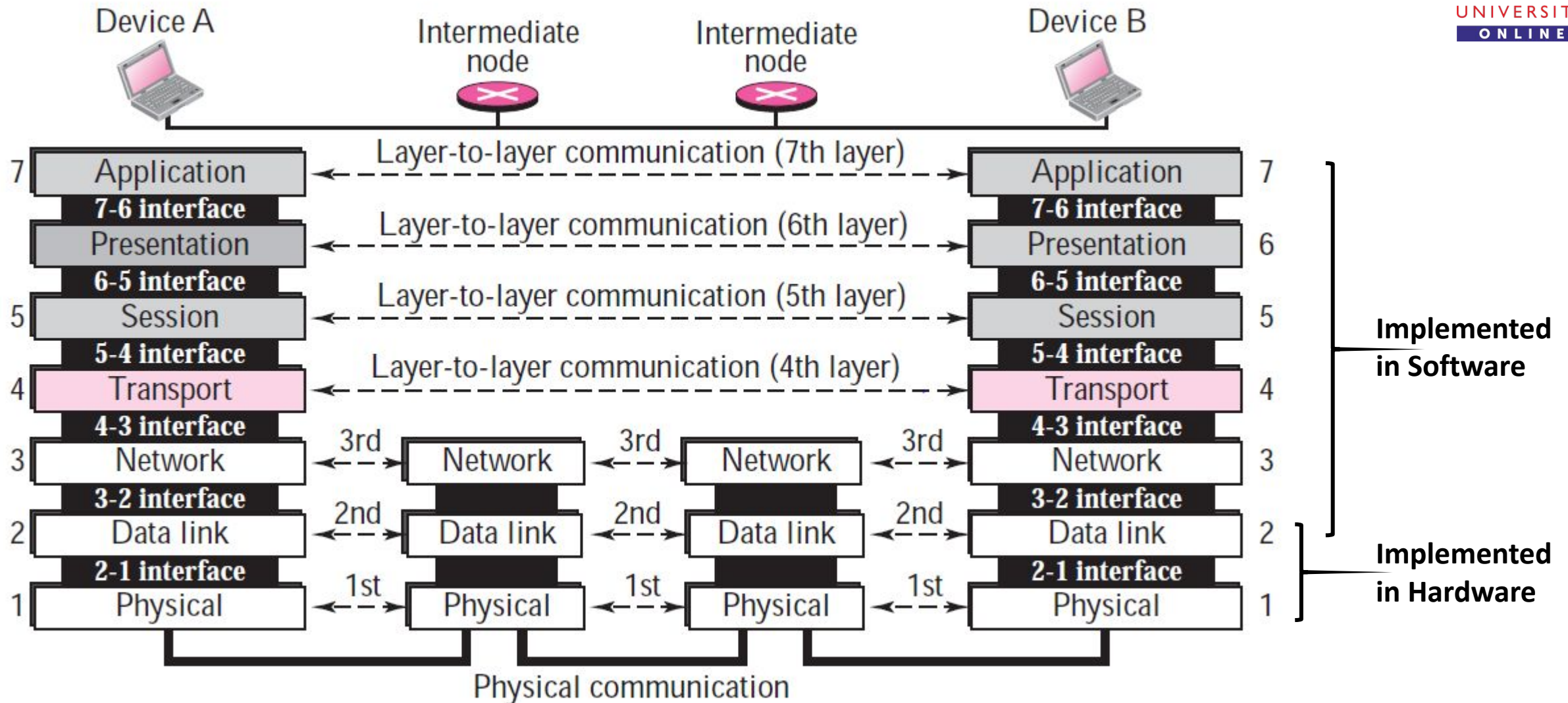
Unit – 4 Link Layer and LAN Roadmap



- Introduction
- Error detection, correction
- Multiple access protocols
- Switched LANs
 - addressing, ARP
 - Ethernet
 - switches
- A day in the life of a web request
- **Physical layer**
- Wireless LANs: IEEE 802.11

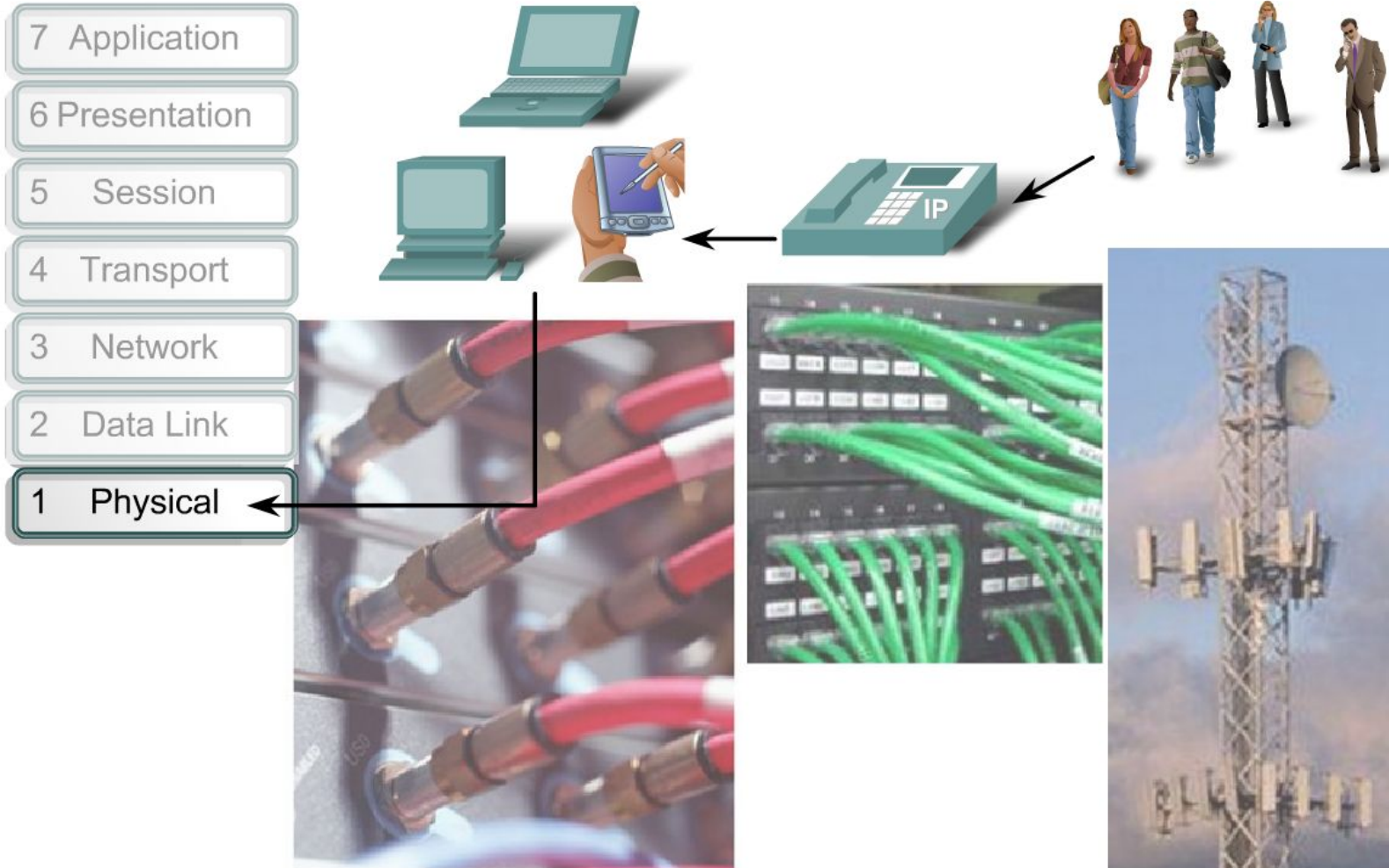
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Physical Layer – OSI Reference Model



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Purpose of Physical Layer



The Physical layer interconnects our data networks.

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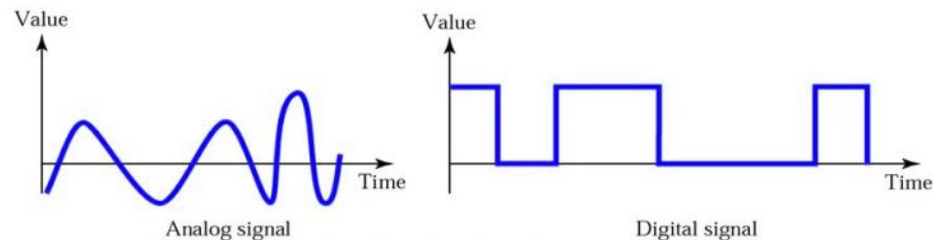
Physical Layer Services



The role of Physical Layer is to:

- encode the binary digits that represent data link layer frames into signals
- to transmit and receive these signals across the physical media
 - copper wires, optical fiber, and wireless that connect network devices.
- Physical medium - conduct a signal in the form of voltage, light, or radio waves from one device to another.

Analog signal –
continuous-valued
(amplitudes, frequencies)
Digital signal – discrete-valued
(0s & 1s)



Media	Signal Type
Copper cable	Patterns of electrical pulses
Fiber-optic cable	Patterns of light pulses
Wireless	Patterns of radio transmissions

The delivery of frames across the local media requires the following physical layer elements:

- The physical media and associated connectors
- A representation of bits on the media
- Encoding of data and control information
- Transmitter and receiver circuitry on the network devices

Components to understand Physical Layer functions includes:

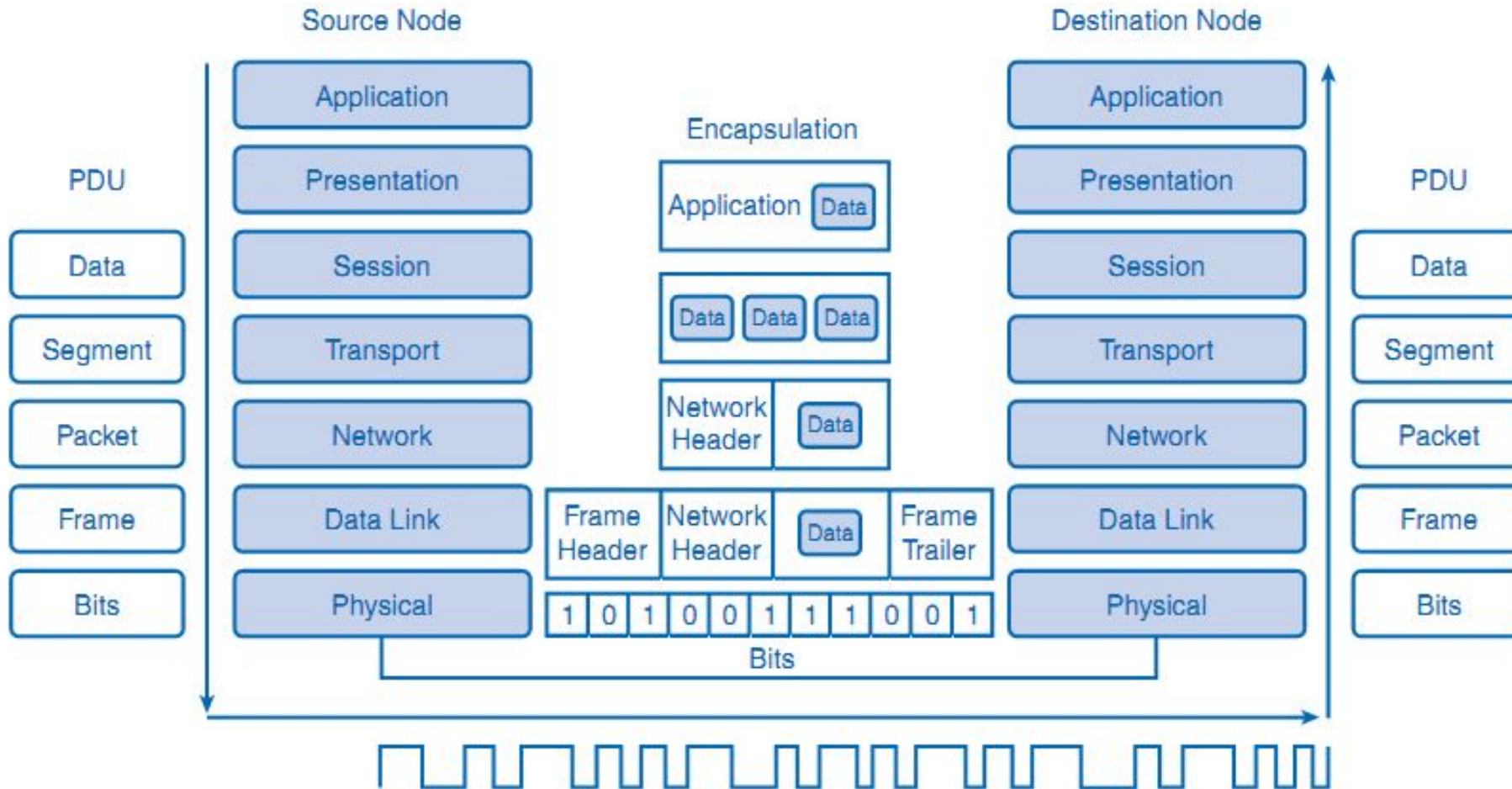
- Physical components
- Encoding
- Signaling

Physical Layer Devices

- Hub
- Repeater
- Modem
- Cables
- Network Adapters

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Physical Layer Services



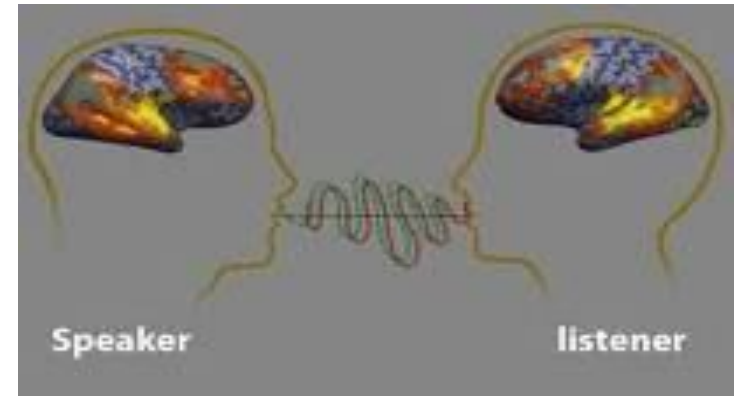
Physical Layer Encoding

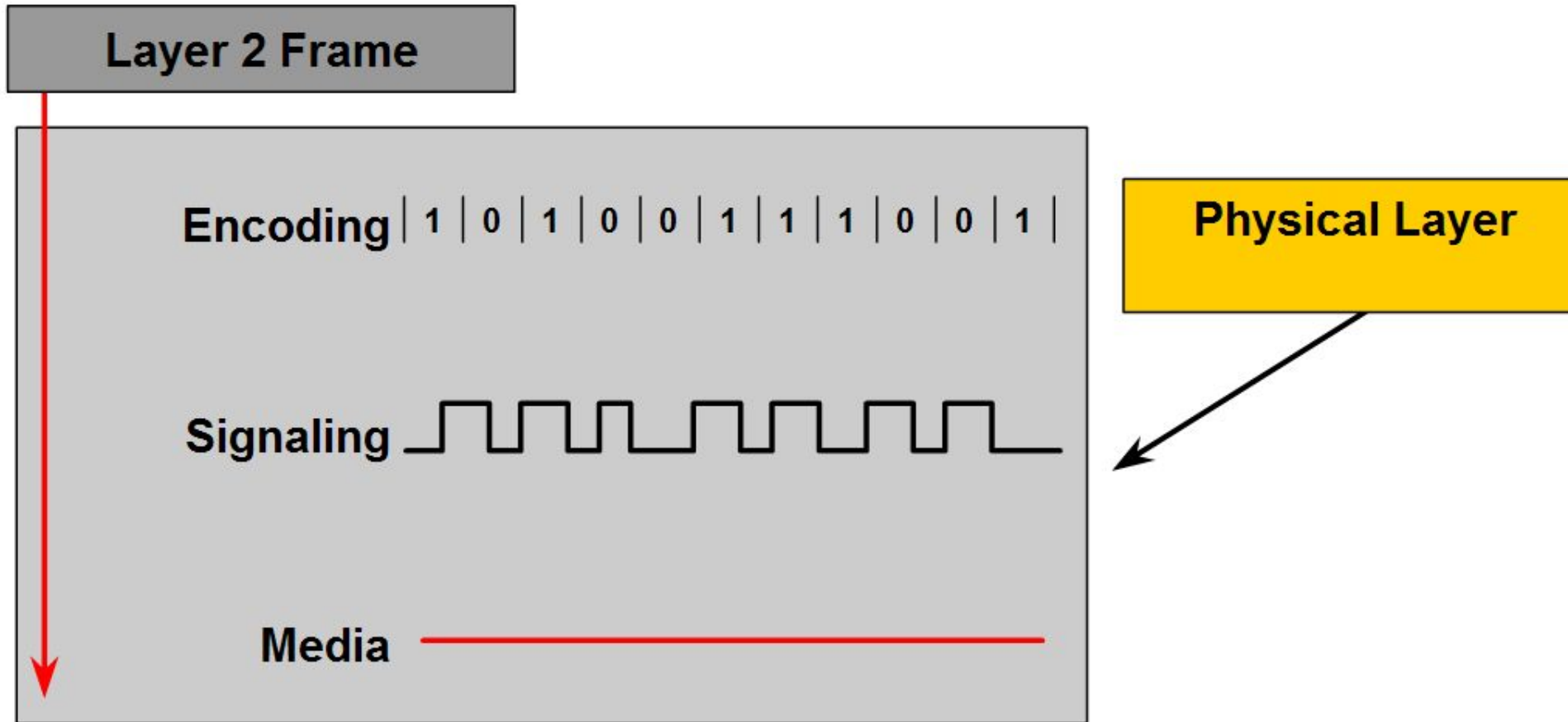
A person wants to communicate an idea:

- processes an abstract thought into words (signaling) ->
- encoded into speech sounds (encoding) ->
- sent out through the medium of air (physical).

At the other end:

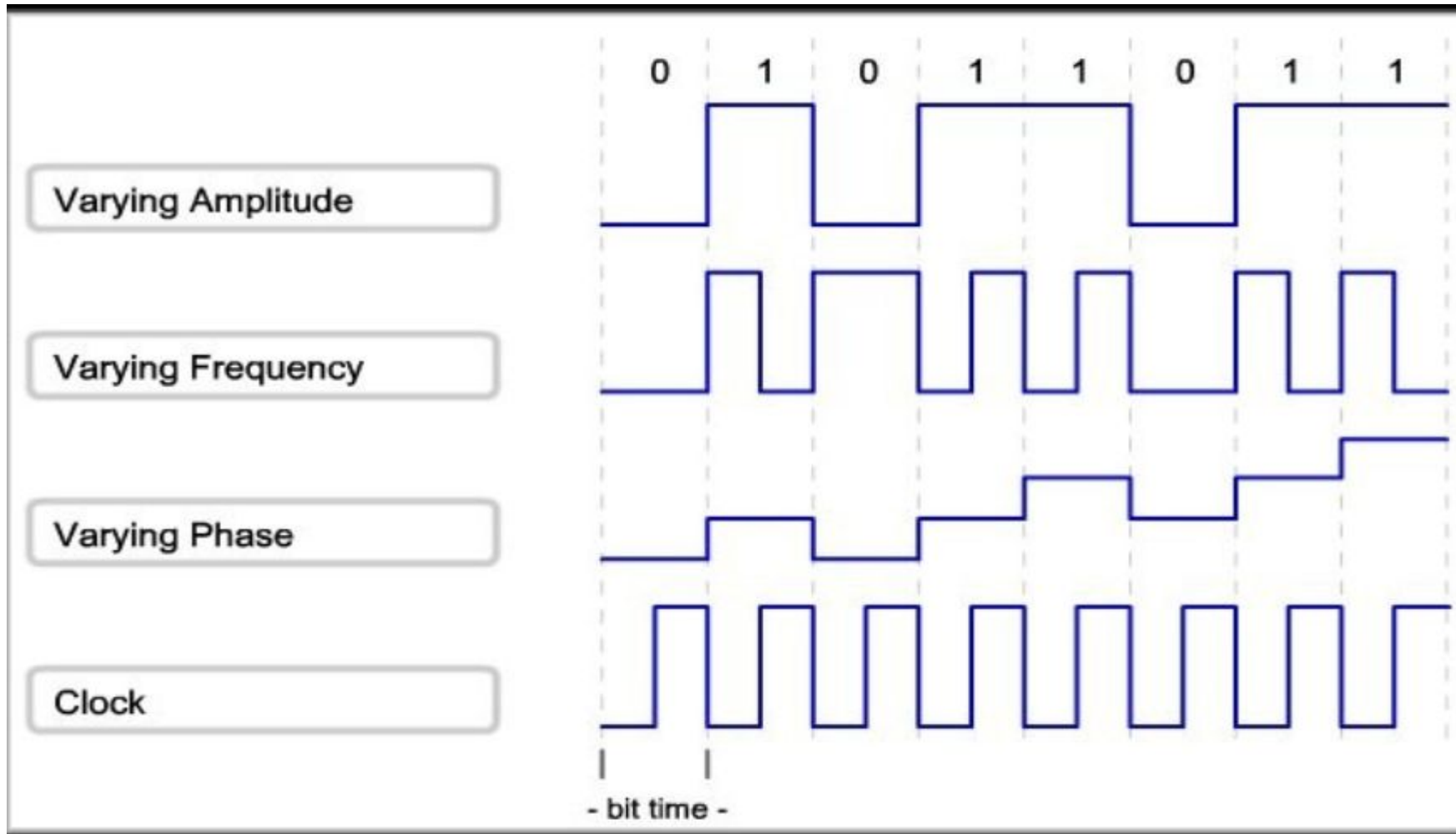
- the receiver interprets the signal of sound,
- recognizes patterns in the sound that denote words &
- then processes the meaning of the words into the original idea





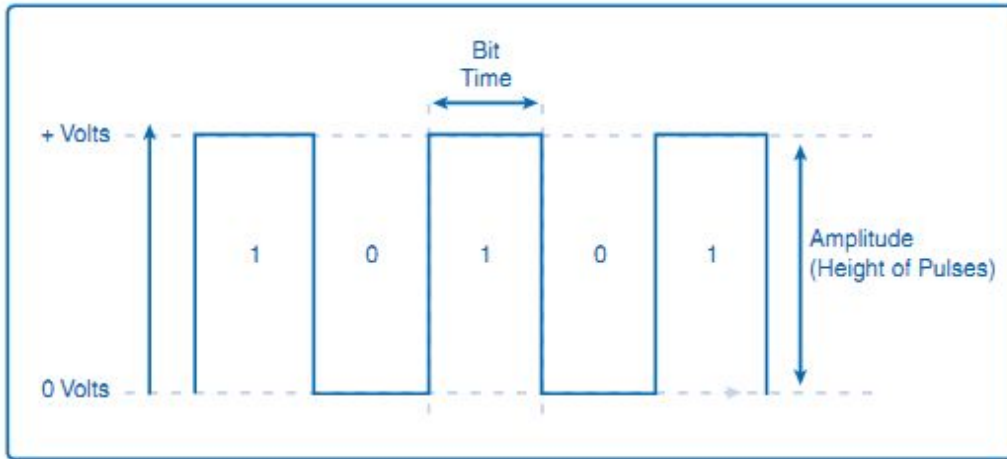
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Signalling Bits for the Media

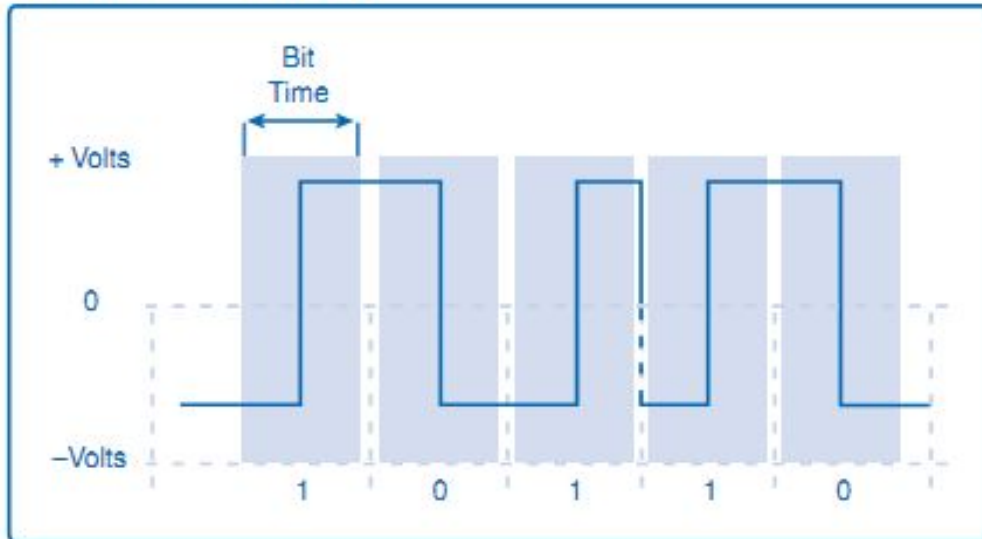


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Signalling Methods



NRZ (Non Return to Zero) Encoding – A low voltage being 0 and a higher voltage representing a 1.

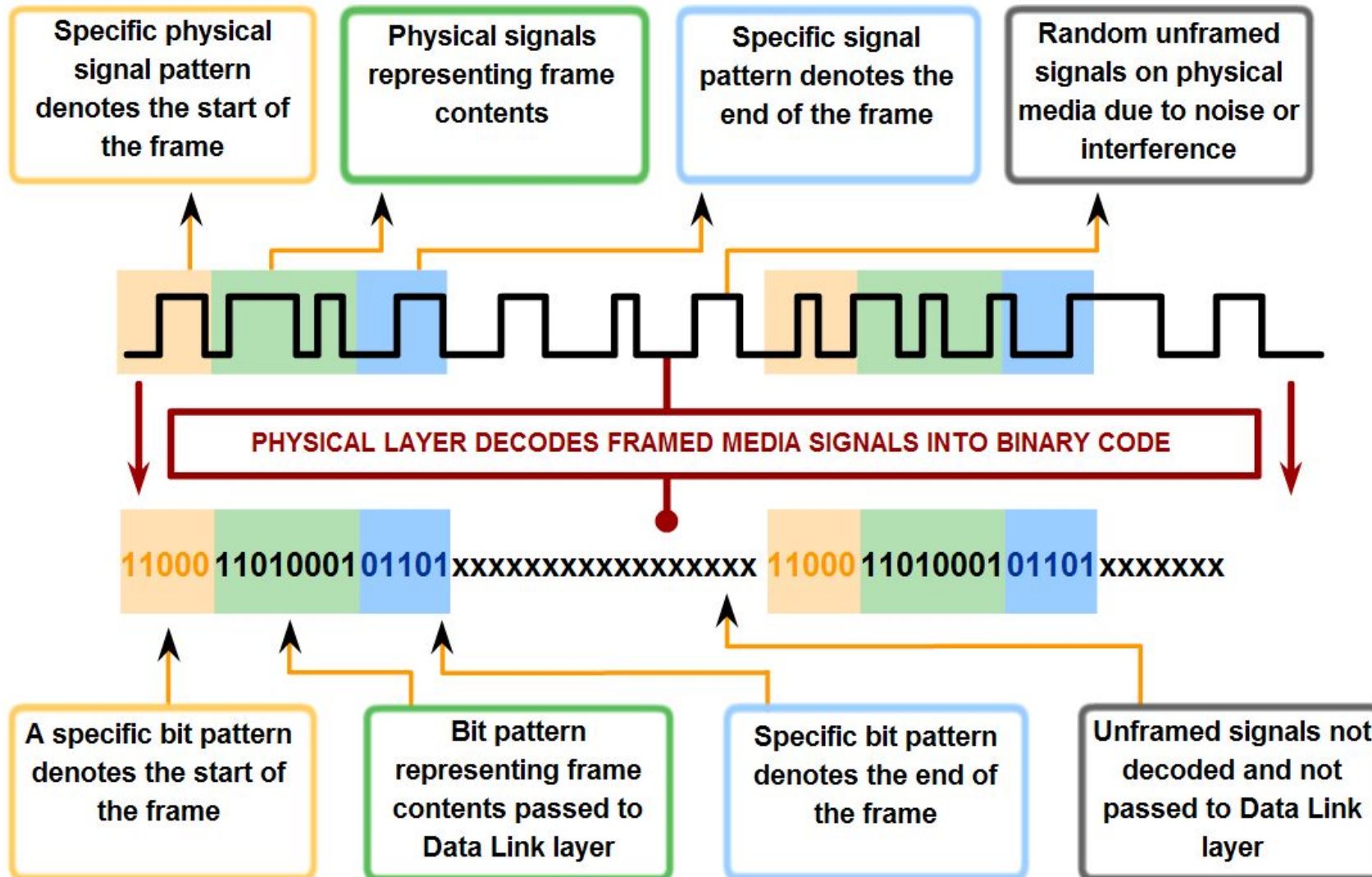


Manchester Encoding

- A voltage change from low to high within the bit time represents a 1.
- A voltage drop within the bit time from a high to a low voltage represents a 0.

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Physical Layer – Signalling and Encoding



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Physical Layer Standards



Some of the key organizations:

- The International Organization for Standardization (ISO)
- The Institute of Electrical and Electronics Engineers (IEEE)
- The American National Standards Institute (ANSI)
- The International Telecommunication Union (ITU)
- The Electronics Industry Alliance/Telecommunications Industry Association (EIA/TIA)
- National telecommunications authorities such as the Federal Communications Commission (FCC) in the United States

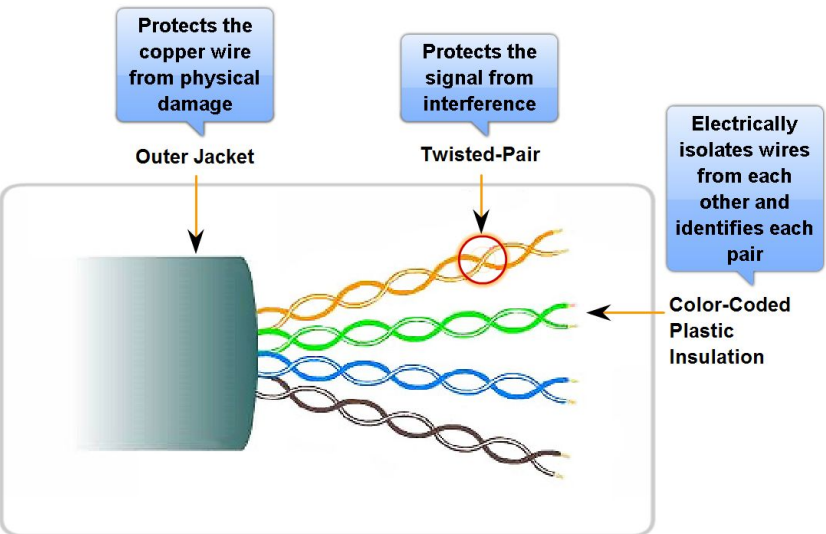
**TCP/IP Standards set by:
IETF**

Standards set by:
ISO IEEE
ANSI ITU
EIA/TIA FCC

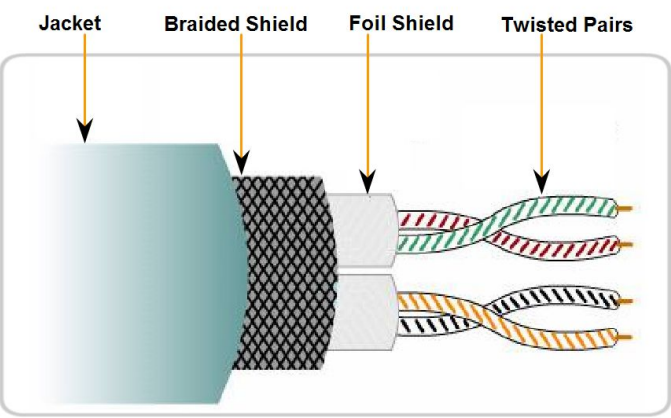
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Transmission Media

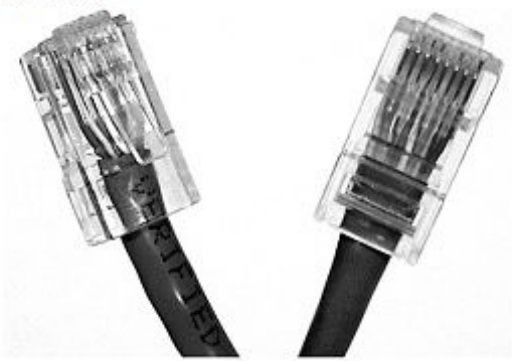
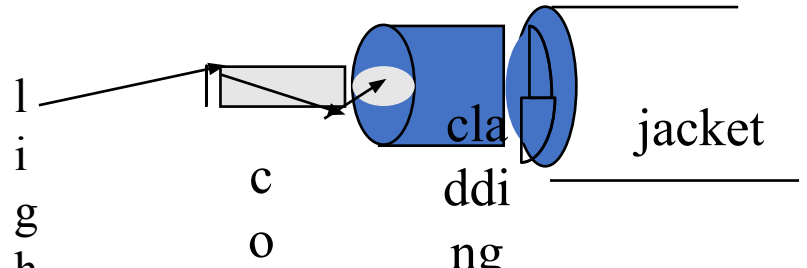
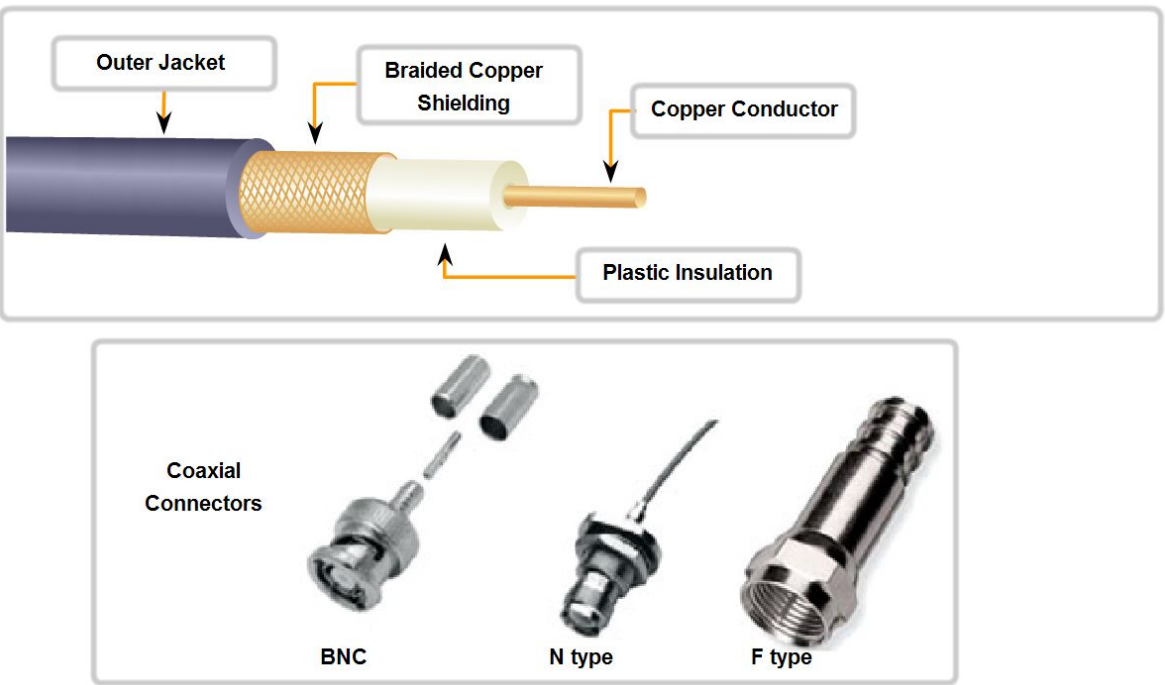
Unshielded Twisted-Pair (UTP) Cable



Shielded Twisted-Pair (STP) Cable



Coaxial Cable Design



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Types of Physical Transmission Media

Specification	Media	Maximum Segment Length	Connector
10BASE-T	CAT 3,4 or 5 UTP (4 pair)	100m	RJ-45
100BASE-TX	CAT 5 UTP (2 pair)	100m	RJ-45
100BASE-FX	62.5/125 multimode fiber	2km	
1000BASE-CX	STP	25m	RJ-45
1000BASE-T	CAT 5 UTP (4 pair)	100m	RJ-45
1000BASE-SX	62.5/50 multimode fiber	62.5 – 275m 50 – 550m	
1000BASE-LX	62.5/50 multimode 9-micron single-mode fiber	62.5/50 – 550m 9 –10 km	
1000BASE-ZX	9-micron single-mode fiber	70km	
10GBASE-ZR	9-micron single-mode fiber	80km	



THANK YOU

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